

Kokulan Kugathanan

 [Kokulan Kugathanan](#) ||  [Kokulan kugathanan](#) ||  kokulankugathanan2003@gmail.com ||  +94 767 520033

SUMMARY

I'm an IT student at SLIIT, passionate about data science and data engineering since starting my journey in 2024. I have hands-on experience in data analysis and machine learning, along with certifications in AI, cyber forensics, and front-end development. I'm currently seeking an internship to enhance my technical skills and contribute to real-world data projects while growing as a collaborative team member.

TECHNICAL SKILL

- **Programming Languages:** Python, Java, JavaScript, C, C++
- **Data Science & AI:** Pandas, NumPy, Scikit-learn, Data Analysis, Machine Learning, Deep Learning, TensorFlow, Keras
- **Web Development:** Django, Spring Boot, React, MERN Stack, REST APIs
- **Databases:** MySQL, PostgreSQL, MongoDB
- **Tools & Platforms:** Git, GitHub, Docker, Hugging Face, Vercel, Render, Cloudinary
- **IoT & Networking:** Arduino, Sensor Integration, Cisco Networking, Network Design
- **Cybersecurity:** Computer Forensics, Ethical Hacking, Security Fundamentals
- **Productivity Tools:** Microsoft 365, Microsoft Copilot, AI for Business

EDUCATION

- **Sri Lanka Institute of Information Technology**
BSc (Hons) in Information Technology, Specialising in Data Science March 2024 - Present
- **J/Kokuvil hindu college** [Jaffna, Sri Lanka.]
High school (Secondary Education) 2015 - 2022
GCE(A/L): Physical science stream

PROJECTS

- **Skin Disease Detection System**
Role: AI / ML Engineer - Self Project
Description: Built a 7-class skin lesion image classifier using a custom CNN and EfficientNet-B0 transfer learning, achieving a macro-AUC of 0.982 and ~88% validation accuracy across 30 epochs with 2-phase fine-tuning. Performed data preprocessing (resizing, normalisation, augmentation, stratified splits), evaluated per-class performance via ROC curves and confusion matrix (NV: 98% recall, DF: 90%, BKL: 94.6%), applied a 0.80 confidence threshold for uncertainty filtering, and deployed the model as a live REST API on Hugging Face Spaces integrated into a web application for real-time predictions.
Technologies: Python, TensorFlow, Keras, OpenCV, NumPy, Pandas, Django/Flask, Hugging Face Spaces
Duration: January - February 2026
Link: <https://huggingface.co/spaces/kokulan123/skin-lesion-api>
- **Medi Connect**
Role: Team Lead & Full-Stack Developer - Group Project
Description: Led the end-to-end design and development of a comprehensive full-stack hospital management platform supporting six user roles (Patient, Doctor, Receptionist, Lab Technician, Cleaning Staff, Admin). Personally developed both the React.js web application and the React Native (Expo) mobile application, delivering role-based responsive dashboards and native mobile features (camera integration, image upload, and text-to-speech). Designed and implemented the backend REST API, JWT authentication, and MongoDB database schema. Integrated an AI-powered diagnostic scanner leveraging a fine-tuned EfficientNet-B0 model to classify 7 different skin diseases from images uploaded via Cloudinary.
Technologies Used: Node.js, Express.js, React.js, React Native (Expo), MongoDB, Mongoose, JWT, Cloudinary, Python, Flask, TensorFlow (EfficientNet-B0), OpenAI API, Vite, Tailwind CSS, Axios
Duration: March – May 2026
Link: <https://github.com/kokulanK/hospital-management-system>, <https://hospital-management-system-three-ebon.vercel.app/>
- **Full-Stack Task Management Application**
Role: Full-Stack Developer – Self Project
Description: Developed a production-ready task management web application with secure JWT-based authentication and role-based access control. Implemented full CRUD operations for task management with priority levels (HIGH/MEDIUM/LOW) and status tracking (TO_DO/IN_PROGRESS/DONE). Performed RESTful API design, database schema modeling, and integrated search and filter functionality. Containerized the entire application using Docker Compose for seamless multi-service deployment.
Technologies Used: Java, Spring Boot, Spring Security, Angular, TypeScript, MySQL, JWT, Docker, Docker Compose, REST API, Hibernate/JPA
Duration: January – March 2026
Link: <https://github.com/kokulanK/task-manager-app>
- **Hospital Management System**
Role: Team Lead & Developer
Description: Led the development of a hospital management web application using Java and the Spring Boot framework. Implemented core CRUD functionalities with robust Object-Oriented Programming (OOP) principles. Applied Data Structures and Algorithms (DSA) concepts, including Bubble Sort for patient record sorting and a Priority Queue for managing appointment priorities. Managed data storage using .txt files to simulate database operations. Designed a modern, responsive user interface with Tailwind CSS.
Technologies Used: Java, Spring Boot, Tailwind CSS, MySQL
Duration: June - November 2025
Link: https://github.com/kokulanK/Medical-Portal-Project_Y2S1

- **Home Automation & Solar Tracking System**

Role: Team Lead & Developer

Description: Designed an Arduino-based IoT project for home automation and solar tracking, integrating LDR sensors, humidity controller, LED display, ultrasonic sensors, and servo motors. The system automates home functions and optimizes solar energy collection by tracking sunlight, inspired by a sunflower model. Completed as part of the IT1140 - Fundamentals of Computing module in Semester 1.

Technologies Used: Arduino, C++, LDR, Humidity Sensor, Ultrasonic Sensors, Servo Motors

Duration: July - October 2024

- **Network Design for TechWorld Inc.**

Role: Team Lead & Network Designer

Description: Led the design and implementation of a network for TechWorld Inc.'s new office, aimed at ensuring secure, scalable, and efficient communication across departments. The project involved developing a VLAN-based structure to separate departmental traffic, setting up core network devices, and securing data with AES-256 encryption. The design includes redundancy, guest network isolation, and provision for future expansion, accommodating a multi-floor, IoT-ready setup.

Technologies Used: Cisco Packet Tracer, AES-256, VLANs, Wireless Access Points

Duration: September - October 2024

CERTIFICATION

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| • SLIIT - AI/ML Engineer – Stage 1 & 2 | May 2026 |
| • Yarl IT Hub Leadership Excellence Program (Cohort 01) | April 2026 |
| • University of Moratuwa - Python for Beginners | December 2025 |
| • University of Moratuwa - Web Design for Beginners | November 2025 |
| • Microsoft Copilot for Productivity by Microsoft and LinkedIn | November 2024 |
| • Cyber Forensics by Great Learning Academy | December 2023 |
| • Front End Development – HTML by Great Learning Academy | December 2023 |
| • Creating a Free Business Page with Blogger – Coursera | December 2023 |

SOFT SKILL

- **Communication:** Able to explain technical concepts clearly and concisely
- **Teamwork:** Collaborating effectively in group projects and community events
- **Problem-Solving:** Logical thinking and troubleshooting skills in project development
- **Adaptability:** Quick learner, eager to take on new challenges and technologies
- **Leadership:** Taking initiative in group projects and as a member of AIESEC and SNUSC
- **Time Management:** Balancing academic, extracurricular, and personal development activities

EXTRA-CURRICULAR ACTIVITIES & LEADERSHIP

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|---|-------------------------------|
| • AIESEC IN SLIIT – TM & IM, IM Lead, Branch Coordinator (Northern Uni) | July 2025 - Present |
| • AIESEC IN SLIIT – OGTA, IR Manager, Team Leader, Member | July 2025 - November 2025 |
| • AIESEC IN SLIIT – Task Force [IGV B2B], Member | September 2024 - January 2025 |
| • SNUSC [SLIIT NORTHERN UNI Student Community] – Member | October 2024 - October 2025 |
| • Workshop Facilitator, Yarl IT Hub(Madu Zone) | May -2025 |
| • Volunteer, Yarl IT Hub Innovation Festival (Maker Space) | August 2-4, 2024 |
| • Carrom Player - Runner-Up, SLIIT Northern University Sports Fest | October 20, 2024 |

LANGUAGES

- **English:** Full Professional Proficiency
- **Tamil:** Native Proficiency

REFERENCES

- **Mr. V. A. M. Ragnathan Sinthujan**

Position: Lecturer, Patron of SNUSC [SLIIT NORTHERN UNI Student Community]

Department: Faculty of Computing | Department of Information Technology

Email: sinthujan.r@slit.lk

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- **Ms. Karuneswaran Gowrisha**

Position: Assistant Lecturer, Year 1 Coordinator [SLIIT Northern University]

Department: Faculty of Computing | Department of Information Technology

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